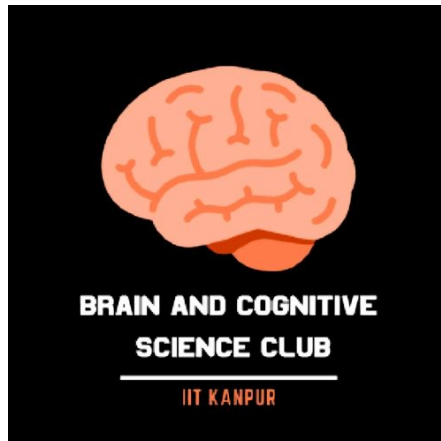




Brain and Cognitive Science Club

Summer Project 2025

**Brain Spy:
Deciphering Alzheimer's**

Final Project Report

Abstract

Brain-Spy is a deep learning framework that employs computer vision methodologies for the automated classification of Alzheimer's Disease progression stages using three-dimensional MRI neuroimaging data. The system utilizes a novel approach that transforms volumetric brain scans into two-dimensional slice representations, subsequently training convolutional neural networks to identify neuroanatomical patterns associated with disease pathology.

The project encompasses a comprehensive pipeline that includes fundamental MRI imaging principles, detailed analysis of Alzheimer's disease and implementation of advanced preprocessing techniques. These preprocessing components include systematic slice extraction, intensity normalization, and skull stripping procedures to optimize data quality for downstream analysis.

The current implementation has achieved successful deployment and evaluation of multiple deep learning architectures for binary classification tasks, demonstrating robust performance in distinguishing between healthy controls and Alzheimer's patients. Additionally, preliminary experimental investigations have been initiated to explore unsupervised learning methodologies for the differentiation of stable Mild Cognitive Impairment (sMCI) from progressive Mild Cognitive Impairment (pMCI) cases, representing an advanced multi-class classification challenge in early-stage disease detection.

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1 Introduction

Alzheimer's Disease (AD) is the most common cause of dementia worldwide, affecting over 55 million people and accounting for 60-70% of dementia cases. Its prevalence is projected to triple by 2050 due to aging populations, making it one of the most pressing health-care challenges of the 21st century. The disease is progressive, irreversible, and currently incurable, gradually eroding memory, cognitive abilities, and independence. Early detection, especially during the preclinical and Mild Cognitive Impairment (MCI) stages, is crucial for timely intervention, improving patient outcomes, and facilitating effective clinical trial recruitment. Studies show that machine learning algorithms are better than clinicians at predicting AD. [1]

1.1 The Role of Neuroimaging in Alzheimer's Detection

Neuroimaging has transformed how we diagnose, study, and monitor AD. Among available imaging modalities—such as Structural MRI, Functional MRI, and Positron Emission Tomography—Magnetic Resonance Imaging (MRI) stands out for its ability to non-invasively capture high-resolution structural details of the brain. MRI can reveal hallmark structural changes in AD, including hippocampal atrophy, cortical thinning, and ventricular enlargement, which often appear years before the onset of severe clinical symptoms.

However, interpreting MRI scans manually is time-consuming, requires specialized expertise, and is subject to variability. Additionally, the structural changes seen in AD can overlap with those from normal aging or other neurodegenerative conditions, making accurate diagnosis challenging.

1.2 Why Deep Learning?

Deep learning—especially Convolutional Neural Networks (CNNs) and related architectures—has revolutionized image analysis by

learning hierarchical, spatially aware features directly from raw data. When applied to 3D brain MRI, deep learning offers the potential to:

- Detect subtle structural patterns not visible to the naked eye.
- Standardize diagnostic assessments across diverse datasets.
- Scale analysis to thousands of scans without manual intervention.

However, the application of deep learning to MRI-based Alzheimer's classification is non-trivial. It faces unique challenges such as:

- **High Dimensionality:** 3D MRI volumes are large, making direct processing computationally expensive.
- **Limited Annotated Data:** Medical datasets are relatively small compared to standard vision datasets like ImageNet, increasing the risk of overfitting.
- **Variability in Acquisition:** Scans differ across sites, scanners, and protocols, introducing unwanted domain shifts.
- **Interpretability:** Clinicians require not only predictions but also insights into *why* the model reached its conclusion.

1.3 Our Work

The limitations of manual diagnosis underscore the need for automated, data-driven approaches to extract reliable biomarker information from MRI data and accurately predict Alzheimer's Disease.

In this work, we utilized a publicly available benchmark dataset and systematically experimented with various preprocessing pipelines, classification architectures, and hyperparameter tuning. Our models achieved accuracies comparable to, and in some cases competitive with the state-of-the-art approaches.

2 Background

2.1 Literature Review

The application of machine learning, particularly deep learning, to the diagnosis of Alzheimer's disease (AD) from magnetic resonance imaging (MRI) has become a prominent area of research. Early studies, such as the work by Klöppel et al. (2008), demonstrated the potential of computer-aided diagnosis by showing that an automated method could achieve accuracy comparable to experienced radiologists. This early work laid the groundwork for the use of computational methods in AD diagnosis. The Alzheimer's Disease Neuroimaging Initiative (ADNI) has been a crucial resource for this research, providing a standardized dataset for algorithm development and comparison. Papers by Jack et al. (2008) and Mueller et al. (2005) detail the MRI protocols and participant demographics of the ADNI cohort, while a review by Weiner et al. (2013) highlights the significant research contributions made using this dataset.

Recent advancements have focused on the use of Convolutional Neural Networks (CNNs) to analyze MRI data. Researchers have explored various CNN architectures to improve diagnostic accuracy. For instance, Folego et al. (2020) employed a 3D CNN, which processes the entire brain volume at once, to classify AD. This approach leverages the spatial context of the entire brain, a significant advantage over methods that analyze slices independently. In contrast, other studies have utilized 2D CNNs, often trained on a slice-by-slice basis. To address the limitations of purely 2D or 3D methods, hybrid approaches have also been developed. For example, Jang and Hwang (2022) introduced the M3T (Multi-plane and Multi-slice Transformer) model, a 2.5D approach that combines information from different anatomical planes (axial, sagittal, coronal) to improve classification performance. This variety of CNN architectures—from 2D to 3D and hybrid 2.5D models—reflects a continuous effort to find the most effective way to repre-

sent and analyze complex brain MRI data.

A significant challenge in deep learning for medical imaging is the "black box" nature of these models, which makes it difficult for clinicians to trust the diagnoses. Addressing this, Lozupone et al. (2024) introduced a novel approach in their "AXIAL" paper, which stands for Attention-based eXplainability for Interpretable Alzheimer's Localized diagnosis. Their work is particularly noteworthy because it combines a 2D CNN-based diagnostic model with an attention-based explainability module. The AXIAL model also generates attention maps that highlight the specific brain regions (such as the hippocampus) that are most influential in the model's decision-making process. This interpretability feature is crucial for gaining clinical acceptance and understanding how these automated systems arrive at their conclusions.

The papers also highlight the use of transfer learning and different training strategies to enhance model performance. Khan et al. (2019) explored transfer learning with intelligent data selection, while Venkatasubramanian et al. (2023) introduced a DHO-based pre-trained ENN model. The work by El-Assy et al. (2024) developed a novel CNN architecture for early AD detection, further demonstrating the ongoing innovations in model design. These studies collectively contribute to the development of more accurate, robust, and interpretable models for AD diagnosis, moving the field closer to clinical application. The combination of foundational research on datasets like ADNI, diverse CNN architectural approaches, and the increasing focus on explainability, as exemplified by the AXIAL paper, illustrates the comprehensive and multi-faceted nature of current research in this domain.

Subject-Wise Splitting While several contemporary studies have demonstrated high accuracy in tasks such as emotion recognition, activity classification, or biometric authentication using deep learning models, a critical limitation in a subset of this literature

is the methodology used for data partitioning. Several papers have reported benchmark-exceeding performance metrics. However, a significant gap in their experimental design is the failure to employ subject-wise data splitting. Instead of ensuring that data from any single participant is exclusively confined to either the training, validation, or test sets, these studies often use a random split, which risks a form of data leakage where the model can learn to recognize specific subject characteristics rather than generalizable patterns. The consequence is an inflated accuracy that does not reflect the model's true performance on unseen individuals. This project addresses this methodological flaw by re-evaluating these models under a rigorous subject-wise splitting protocol, thereby providing a more robust and realistic assessment of their generalization capabilities.

2.2 ADNI Dataset

ADNI Initiative and Phases The Alzheimer's Disease Neuroimaging Initiative (ADNI) is a large, multi-site longitudinal study launched in 2004 to develop and validate biomarkers for tracking Alzheimer's disease (AD) progression [2, 3]. ADNI's primary objectives include standardizing MRI and PET imaging protocols, collecting cognitive and clinical measures, and establishing an open data repository for research on normal aging, mild cognitive impairment (MCI), and AD. Over successive phases (ADNI-1 through ADNI-4, spanning 2004–2027), the study enrolled cohorts resembling clinical trial designs (approximately 40% cognitively normal (CN), 40% MCI, and 20% mild AD) [4]. For example, ADNI-1 (2004–2010) recruited ~819 elderly participants (229 CN, 398 MCI, 192 AD) followed longitudinally. Subsequent phases (ADNI-Go, ADNI-2, ADNI-3, ADNI-4) expanded the cohort and modalities, adding early MCI arms, tau PET imaging, and remote assessments. ADNI employs rigorous quality control (QC) on imaging data, enabling reproducible research, and has been cited in over 5,500 scientific publications.

ADNI-1: Complete 1-Year 1.5T Subset

This work focuses on the ADNI-1 *Complete 1-Year 1.5T* subset, defined as all ADNI-1 participants who underwent 1.5 Tesla MRI scans at baseline, 6 months, and 12 months [5]. These are high-resolution T1-weighted structural scans (3D MPRAGE) acquired at multiple sites (GE, Philips, Siemens scanners) following a harmonized protocol. The subset contains ~2000 brain MRI volumes from roughly 600–650 individuals, each with up to three longitudinal scans. After QC, approximately 1,891 scans remain usable. The cohort includes CN, MCI, and AD subjects, with baseline composition around 28% CN, 49% MCI, and 23% AD. Within MCI, *stable MCI* (sMCI) subjects remain MCI over follow-up, while *progressive MCI* (pMCI) subjects convert to AD within ~3 years. For example, one cohort had 68 pMCI and 112 sMCI subjects.

Relevance for AD/CN and MCI Progression Studies

The ADNI-1 *Complete 1-Year 1.5T* dataset is widely used for classification and prognostic modeling in Alzheimer's research. Its standardized longitudinal MRI and clinical data make it ideal for machine learning studies distinguishing AD vs CN and predicting MCI-to-AD conversion. Studies have reported classification accuracies above 90% for AD vs CN and around 75% for pMCI vs sMCI using MRI-derived features [6]. The dataset is a benchmark in public challenges and a reference for algorithm development.

Challenges and Limitations Despite its strengths, the dataset presents challenges:

- **Class imbalance:** CN, MCI, and AD groups are uneven, and sMCI generally outnumbers pMCI. This imbalance can bias classifiers without proper handling (e.g., resampling, class weighting).
- **Missing data:** Not all subjects have every modality or timepoint. CSF data are incomplete, and some participants drop out before 12 months.

- **Cohort bias and heterogeneity:** The sample is predominantly White and highly educated, limiting generalizability. Scanner and site differences introduce residual variability despite harmonized acquisition.

3 Experimentation

In this section, we first analyze different preprocessing methods commonly used in the area prior to finalizing a definitive pipeline. Subsequently, we delve into the implementation of several existing models.

All models were implemented on Kaggle using hardware specifications - 2 x NVIDIA T4 GPUs and 29GiB of RAM.

3.1 Pre-Processing

Preprocessing is essential in deep learning-based Alzheimer's detection as it minimizes acquisition variability, enhances signal-to-noise ratio, and standardizes spatial and intensity features. This ensures models focus on disease-specific anatomical changes, improving training stability, generalization, and reproducibility across diverse datasets.

3.1.1 Pre-Processing Techniques:

Resampling using Torchio (Spatial Normalization): Torchio is an open-source Python library designed for preprocessing MRI images. Resampling modifies the voxel spacing of images, producing standardized input dimensions across subjects. This step is crucial for ensuring spatial consistency, which improves training stability and performance of deep learning models. It eliminates discrepancies in physical space that could arise from varying acquisition protocols, enabling models to generalize better across datasets. By enforcing a common voxel resolution, resampling ensures that anatomical features are represented on the same spatial scale, which is particularly important for convolutional architectures.

Canonical Orientation (ToCanonical): ToCanonical converts medical images to a standard orientation defined by the RAS (Right–Anterior–Superior) coordinate system. Different scanners and acquisition setups may store images in inconsistent axis orders, which can confuse spatially-aware models. This transformation ensures that all images align consistently in terms of anatomical direction. The process involves reordering axes and adjusting affine metadata without distorting the image content. By using a canonical orientation, the preprocessing pipeline guarantees that downstream analyses interpret the same anatomical structures in the same spatial context across the dataset.

Bias Field Correction: Bias field correction removes smooth, low-frequency intensity variations in MRI scans caused by scanner imperfections or coil sensitivity differences. These artifacts can obscure true tissue contrast and mislead analysis. By estimating and compensating for the bias field, this step produces uniform intensity distributions, improving tissue contrast, segmentation accuracy, and reducing scanner-specific artifacts.

Skull Stripping: Skull stripping is a preprocessing step in medical image analysis that removes non-brain tissue such as fat, skull, scalp, and eyes from the image. These surrounding structures can introduce noise, irrelevant features, and unwanted gradients, negatively affecting learning. By eliminating them, skull stripping isolates the brain and improves signal-to-noise ratio, focusing computational resources on the relevant anatomy. This step is essential in tasks like brain segmentation or lesion detection, where preserving only brain matter reduces overfitting and improves model interpretability and performance.

CropOrPad: CropOrPad is a Torchio operation that standardizes image dimensions by either cropping the edges or padding with zeros. Voxels near the image borders are removed,

and if the target shape is larger, zero-valued voxels are added to fill the volume. This ensures all images have consistent input sizes, which is required for batching and model input constraints. Zero-padding preserves spatial alignment without distorting anatomical features, while cropping focuses on the central structures, typically the brain. This transformation simplifies downstream processing and prevents shape mismatches during model training.

CLAHE (Contrast Limited Adaptive Histogram Equalization): CLAHE is an adaptive histogram equalization technique that enhances local contrast in medical images. The image is divided into small, disjoint chunks, and histogram equalization is applied to each, improving the visibility of fine-grained structures. CLAHE limits the contrast amplification to prevent noise over-enhancement, making it especially useful when lighting or intensity distribution varies across the image. This method is effective in revealing subtle anatomical differences and improving downstream model performance in tasks where edge clarity and regional intensity variation are important.

Z-Normalization: Z-normalization is an intensity standardization technique that adjusts the image to have a mean of 0 and a standard deviation of 1. Since MRI intensity values vary across scans and subjects due to acquisition parameters, this normalization ensures that all inputs follow the same distribution. It helps neural networks converge faster and prevents the model from learning scanner-specific features. Standardizing intensities across volumes allows the model to focus on structural and spatial patterns, which are consistent indicators of the condition being studied or diagnosed.

Data Augmentation: Data augmentation introduces controlled variations into training data to improve generalization. Techniques include random affine transformations (translation, rotation, scaling), elastic deformation, flipping, anisotropy (resolution variation across axes), noise addition, blurring, and gamma correction. These augmentations simulate realistic variability in imaging conditions and anatomical presentation while preserving diagnostic content. By diversifying training samples, augmentation prevents overfitting and improves robustness. It is particularly important in medical imaging, where annotated data is scarce, and models must perform reliably across slight variations in input data.

3.1.2 Final Pre-Processing Pipeline:

While Skull-Stripping is very important process in pre-processing, usual toolkits used for skull-stripping are quite computationally expensive. Considering the large size of ADNI dataset, we needed a fast and efficient method for pre-processing.

Neural Pre-Processing (NPP) model: Neural Pre-Processing (NPP) is a learning framework for End-to-end Brain MRI pre-processing. NPP jointly performs intensity normalization, skull stripping, and spatial normalization, while concurrently estimating the bias field to facilitate bias-field correction.

In comparative experiments, NPP demonstrated a substantial improvement in computational efficiency over state-of-the-art methods. Specifically, when evaluated on an NVIDIA GeForce RTX 4060 GPU with 8 GB VRAM, NPP completed all pre-processing steps in approximately 6 seconds, compared to 55 seconds for the widely used SynthStrip method, yielding an approximate 9-fold acceleration.

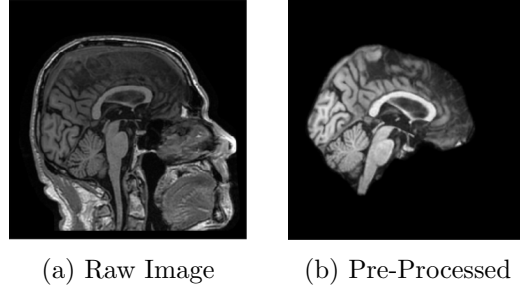


Figure 1: Final Pre-Preprocessing pipeline

<i>Method</i>	<i>SS</i>	<i>IN</i>	<i>SN</i>	<i>GPU</i>	<i>CPU</i>
SynthStrip	16.5	-	-	✓	
BET	9.1	262.2	-		✓
Freesurfer	747.2	481.5	671.6		✓
NPP (Ours)	-	2.94	-	✓	

Table 1: Reported runtimes in NPP paper. Supported sub-tasks and average runtime for each method. Skull-stripping (SS), intensity normalization (IN) and spatial normalization (SN). Units are in seconds

3.2 Models

3.2.1 Pure Transfer Learning Based Approaches

These sets of approaches employ standard transfer learning with slight modifications in the fully connected layers to classify the 3D MRI scans. The work by passing each slice of a 3D scan through the CNN and then classifying each scan by using combination of the results of the classification of those slices. Almost all the conventional ImageNet Large Scale Visual Recognition Challenge (ILSVRC) winning models were tried for this approach. [7]

Resnet152: All but the last 5 layers (excluding the fully connected ones) were frozen and three fully connected layers (each with 4096 units) are added, followed by a sigmoid output for binary classification.

VGG19: VGG19 base model is used with all layers frozen along with the same 3 x 4096 units dense layer setup that was used above.

LeNet and AlexNet: These were also trained with similar setups as above.

3.2.2 DHO Based Model

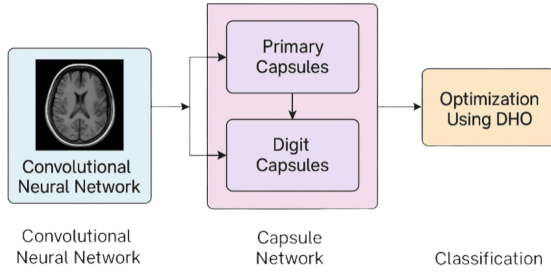
This method combines a pretrained Convolutional Neural Network (CNN), a Capsule Network (CapsNet), and the Deer Hunting Optimization (DHO) algorithm to improve classification performance on MRI data. [8] For this model, 2D axial slices of 224 x 224. The model architecture is as follows:

- **Capsule Network Integration:** The feature maps from the CNN are passed to a Capsule Network, which includes:
 - **Primary Capsules:** These capture low-level features using convolutional capsule layers followed by a squashing nonlinearity.
 - **Digit Capsules:** These high-level capsules aggregate information using dynamic routing, representing entire objects or classes (e.g., AD, NC).

CapsNet preserves hierarchical relationships and spatial orientation, making it particularly suited for brain imaging analysis.

- **Classification:** The output vector lengths from the Digit Capsules represent the probability of each class. The final prediction is obtained by selecting the class with the longest capsule vector.
- **Optimization Using DHO:** The Deer Hunting Optimization (DHO) algorithm, inspired by cooperative hunting behavior, is applied to:
 1. **Feature Selection:** DHO selects the most discriminative features from the CNN-CapsNet output, reducing dimensionality and noise.
 2. **Hyperparameter Optimization:** DHO tunes key model parameters including learning rate, number of epochs, batch size, and CapsNet routing iterations.

This enhances the model's generalization and reduces overfitting.



3.2.3 M3T

The M3T (Multi-plane and Multi-slice Transformer) is a hybrid 3D medical-image classifier that combines a lightweight 3D CNN, multi-plane / multi-slice 2D CNN processing (with ImageNet pretraining), learnable positional/plane embeddings, and a Transformer encoder to capture long-range dependencies across slices and planes. The overall design and block definitions follow Jang & Hwang.

- **Input preprocessing.** 3D T1 MRI volumes are preprocessed (bias correction,

skull stripping, resampling to 128^3 , intensity normalization) and fed into the network as $I \in \mathbb{R}^{L \times W \times H}$ (with $L = W = H = 128$ in the paper).

- **3D CNN block (local 3D features).** A shallow 3D CNN D_{3d} (two $5 \times 5 \times 5$ conv layers with BN+ReLU) produces a 3D representation $X = D_{3d}(I) \in \mathbb{R}^{C \cdot 3d \times L \times W \times H}$ that preserves spatial resolution and encodes local volumetric structure
- **Multi-plane / multi-slice extraction.** From X the model extracts slice feature stacks along the three anatomical planes via operators E_{cor}, E_{sag}, E_{ax} , producing coronal, sagittal and axial slice feature sets: $S = [S_{cor}, S_{sag}, S_{ax}]$, where each S has shape (channels $\times$ slices $\times$ spatial_H $\times$ spatial_W). These are reshaped into a sequence of 2D-like images (tokens).
- **2D CNN + projection (token formation).** Each extracted slice is processed by a shared, weight-tied 2D CNN (ResNet50 pretrained on ImageNet in the paper) to obtain per-slice feature vectors $K \in \mathbb{R}^{3N \times C \cdot 2d}$. A two-layer MLP projection (nonlinear projection) maps these to fixed-dim tokens: $T = D_{mlp}(D_{2d}(S)) \in \mathbb{R}^{3N \times d}$. These tokens bring the benefit of 2D pretraining while still originating from native 3D features.
- **Positional and plane embeddings + special tokens.** Learnable 1-D positional embeddings P_{pos} and plane embeddings P_{pln} are added. A classification token z_{cls} is prepended and plane-separator tokens z_{sep} are inserted between plane groups. The initial transformer input is: $Z_0 = [z_{cls}, T_{cor}, z_{sep}, T_{sag}, z_{sep}, T_{ax}, z_{sep}] + P_{pos} + P_{pln}$. (Notation and sizes follow the paper.)
- **Transformer encoder (global relationships).** The token sequence Z_0 is fed to a stack of K Transformer encoder

layers (MSA + MLP + LayerNorm residual blocks) to model long-range interactions across slices and planes. The paper uses $K = 8$, hidden/MLP sizes and multihead settings ($d=256$, hidden=768, heads=8). The final classification is produced from the transformed classification token via an MLP head.;

3.2.4 ADNet 3D

ADNet3D Network: A 3D convolutional neural network was implemented with a feature extraction backbone consisting of five convolutional blocks. This architecture was referenced from [9].

- **Block 1:** Conv3d($1 \rightarrow 32$, kernel=3, padding=1) + BatchNorm3d + ReLU + MaxPool3d(2)
- **Block 2:** Conv3d($32 \rightarrow 64$, kernel=3, padding=1) + BatchNorm3d + ReLU + MaxPool3d(2)
- **Block 3:** Conv3d($64 \rightarrow 128$, kernel=3, padding=1) + BatchNorm3d + ReLU + MaxPool3d(2)
- **Block 4:** Conv3d($128 \rightarrow 256$, kernel=3, padding=1) + BatchNorm3d + ReLU + MaxPool3d(2)
- **Block 5:** Conv3d($256 \rightarrow 256$, kernel=3, padding=1) + BatchNorm3d + ReLU + MaxPool3d(2)

Classifier Head: A two-layer fully connected classifier with 512 hidden units was used. Layer normalization, ReLU activation, and dropout ($p = 0.5$) were applied to prevent overfitting. The final output layer maps to three classes (CN, MCI, AD).

3.2.5 Dual-Branch 3D CNN

This model employs a unique dual-branch architecture designed to capture complementary feature representations:

- **Branch 1: Fine-grained Feature Extraction**

- Utilizes $3 \times 3 \times 3$ convolutional kernels.
- Captures local, detailed anatomical features.
- Architecture: $1 \rightarrow 16 \rightarrow 64 \rightarrow 256$ channels.
- Applies global average pooling followed by a fully connected (FC) layer reducing the features from 256 to 128 dimensions.

- **Branch 2: Coarse-grained Feature Extraction**

- Utilizes $5 \times 5 \times 5$ convolutional kernels.
- Captures broader spatial patterns and contextual information.
- Architecture: $1 \rightarrow 32 \rightarrow 128 \rightarrow 512$ channels.
- Applies global average pooling followed by a fully connected (FC) layer reducing the features from 512 to 128 dimensions.

- **Fusion Layer**

- Concatenates features from both branches, resulting in a combined feature vector of 256 dimensions.
- Employs final classification layers structured as $256 \rightarrow 128 \rightarrow 64 \rightarrow 2$ classes.

3.2.6 Dual CNN: Single-View Model (Axial Slice Only)

In the initial approach, only the central axial slice (a horizontal cross-section) of the 3D MRI scan was used. This single 2D image was passed through a double-branch CNN (ADNet), composed of two different convolutional pathways: [10]

- **CNN1:** Small receptive field (3×3) to capture localized anatomical features.

- **CNN2:** Larger receptive field (5x5) to capture global neurodegenerative patterns.

The outputs of both CNNs (each producing a 128-dimensional vector) were concatenated to form a 256-dimensional embedding, which was passed through fully connected layers for classification.

CNN1 Architecture: CNN1 focuses on fine-grained spatial features using smaller receptive fields:

- Two 3x3 Conv2D layers (input to 16 channels), ReLU activations, and 2x2 MaxPooling.
- Two more Conv2D layers expanding to 64 channels, followed by ReLU and MaxPooling.
- Final Conv2D expands to 256 channels, followed by MaxPooling.
- Output tensor of shape (256, 32, 32) is flattened to a 262144-dimensional vector.
- Fully connected layer reduces to 128-dimensional feature vector.

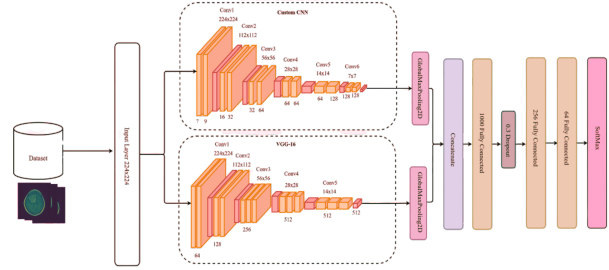
CNN2 Architecture: CNN2 captures broader structural patterns using wider filters:

- Begins with Conv2D layers with 5x5 kernels, mapping from 1 to 32 channels.
- Intermediate layers expand to 128 and 512 channels with ReLU and MaxPooling.
- Output tensor of shape (512, 32, 32) is flattened to a 430592-dimensional vector.
- Fully connected layer reduces it to 128-dimensional representation.

TwinCNN Module: TwinCNN accepts a 2D image slice and processes it through both CNN1 and CNN2:

- CNN1 extracts local features; CNN2 captures global patterns.

- Each produces a 128-dimensional vector.
- These vectors are concatenated into a 256-dimensional feature vector.



3.2.7 AXIAL

This model aims to capture and use the relationship between multiple slices by using an Attention XAI Fusion Module and create a final representation of that particular MRI scan and then use it for classification. Basically, this model tries to use global features for an entire MRI scan rather than classify each 2D slice by tagging it with its parent label. [11].

- **Feature Extraction Module:**

- Input: N individual 2D brain slices (e.g., axial MRI slices from a 3D volume).
- Process: Each slice is passed through a CNN backbone (VGG16).
- Output: A feature tensor of shape (N, f_d) — where each slice is represented by a vector of length (f_d) (after the flattening the output of VGG).

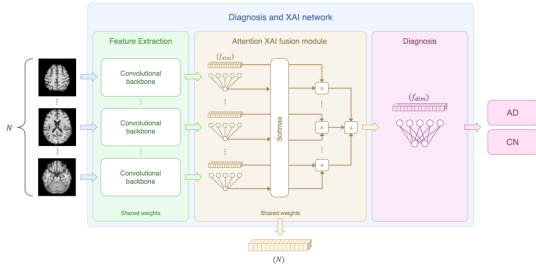
- **Attention XAI Fusion Module:**

- Purpose: Aggregate slice-wise features into a single subject-level representation with explainability.
- Steps:
 - * Each feature vector (f_d) is passed through a small MLP to compute attention scores.

- * A softmax is applied over the scores.
- * Each feature is multiplied by its attention weight.
- * A weighted sum of all features produces a single vector.
- Output: A single fused feature vector representing the full subject's brain scan.

• Diagnosis Module:

- Input: The fused feature vector from the attention module
- Process: Fed into a fully connected MLP classifier.
- Output: Labels corresponding to AD/CN.



4 Methods

4.1 Modified AXIAL

Our method closely resembles the AXIAL approach mentioned above but introduces some tweaks which cause it to fare better than the former. The model consists of three modules, **Input, AttentionXAI Fusion, Classification**. The input module is responsible for handling the inputs and passing them through the Convolution base for feature extraction. The AttentionXAI Fusion module is responsible for calculating the attention scores, and the final feature vector F for each brain scan. The final module is a simple MLP that produces logits for the input and helps in classification. In the following, B is the batch size and N is

the number of slices per scan, i.e., 100 in our case. N is selected using the entropy-based selection [7]. By this mechanism, we end up with roughly 118,200 2D slices in our design matrix, X , which consists of 128x128 center crops from each slice and stacked channels to maintain consistency with further layers of the model.

• Input Module:

- The input is processed and is fed to the model as a tensor of shape $(B, N, 3, 128, 128)$.
- A ResNet152 convolution backbone (the top layers, i.e., the fully connected ones are removed and the model is frozen) is applied to the input and the outputs are feature maps of shape $(B, N, 2048)$ after flattening.

• Attention XAI Fusion Module:

- The output of the previous module is fed as input into this module while retaining the individual feature vectors for each slice. Now, we use a linear transformation to map this high dimensional tensor into a lower dimension (128). This is the weight matrix, W_1 and the output of this step is a tensor of shape $(B, N, 128)$.
- A $\tanh()$ activation is now applied on this tensor throughout.
- Now, another linear transformation, the matrix W_2 is applied to map the 128 dimensional vector to a single scalar value i.e., the attention score α for each slice. The tensor is now shaped $(B, N, 1)$ which after squeezing becomes (B, N) . A softmax is also applied to normalize these attention scores and a subsequent un-squeezing occurs which gives us normalized attention scores in the shape $(B, N, 1)$.

- Now we wish to calculate the feature vector for each scan F , which is given as $F = \alpha f$. And it can be seen that F is a tensor of shape $(B, 2048)$. We have now successfully mapped the entire brain scan to a 2048 dimensional column vector.

- **Classifier Module:**

- The output vectors F are now fed into a simple MLP to compute the logits for each scan.
- We first map F to a 128 dimensional vector, then apply a ReLU activation, then a dropout layer and finally another linear transformation to produce the logits. A sigmoid is also applied at train-test time to these raw logits.

The Adam Optimizer was used for this module with a learning rate of $1e-4$. 72% of the dataset was used for training, 8% for validation, and 20% for testing.

4.2 Dual CNN: Multi-View Weighted Model (ThreeHeadedADNet)

This is a modification of the Dual CNN model mentioned above. This approach introduces anatomical diversity by processing the central slices from all three axes: axial, coronal, and sagittal. Each slice was passed through an identical copy of the double-branch CNN (TwinCNN). To improve decision-making, a learnable attention mechanism was added:

- Each anatomical view produced a 256-dimensional vector.
- A trainable weight vector (α) was applied to the three outputs using a softmax function.
- The weighted features were concatenated to form a 768-dimensional vector, passed through dense layers for final classification.

Each 3D MRI scan was decomposed into three distinct 2D slices from different anatomical planes:

- **Axial slice (X1):** horizontal cross section, taken along the third axis.
- **Coronal slice (X2):** front-back view, taken along the second axis.
- **Sagittal slice (X3):** left-right side view, taken along the first axis.

Each slice is converted to a PyTorch tensor of shape $(1, H, W)$, representing grayscale images with channel-first format. This preprocessing mirrors clinical practice, where radiologists examine multiple orientations for a comprehensive analysis.

W_ThreeHeadedADNet Architecture

W_ThreeHeadedADNet uses three parallel TwinCNN branches, one for each anatomical view:

- Each branch (head1, head2, head3) processes one of the slices (X1, X2, X3).
- Outputs are 256-dimensional vectors.
- A trainable attention vector $\alpha = [\alpha_1, \alpha_2, \alpha_3]$ (via softmax) weights each head:

$$h_1 = \alpha_1 \cdot \text{head1}(x_1), \quad h_2 = \alpha_2 \cdot \text{head2}(x_2),$$

$$h_3 = \alpha_3 \cdot \text{head3}(x_3)$$

- Weighted outputs are concatenated to form a 768-dimensional feature vector.
- Final classifier:
 - Linear(768 $\rightarrow$ 256) + ReLU + Dropout
 - Linear(256 $\rightarrow$ 2) for binary classification

Table 2: Results and Comparison of Different Models

Model Name	Accuracies			F1 Scores		
	Train	Validation	Test	F1 (AD)	F1 (CN)	Macro F1
Modified AXIAL (Ours)	0.9859	0.9662	0.9700	0.960	0.970	0.965
Dual CNN: Multi View (Ours)	1.0000	0.9195	0.9667	0.9655	0.9677	0.9667
AXIAL	-	-	0.8560	-	-	0.712#
Dual CNN (Single View)	1.0000	0.9492	0.9500	0.9492	0.9508	0.950
Dual 3D CNN	-	0.7904	0.8087	0.6800	0.8600	0.770
ADNet3D	0.9575	0.6404	-	-	-	0.708**
3D CNN (Pretrained)	-	0.4845	-	-	-	-
ResNet (T-Learning)	1.0000*	0.9669*	0.9632*	-	-	-
VGG19 (T-Learning)	0.9886*	0.9354*	0.9349*	-	-	-
AlexNet (T-Learning)	-	-	0.5900	0.0700	0.5600	0.315
LeNet (T-Learning)	-	-	0.3800	0.4100	0.3300	0.370
ViT	0.4498	0.5838	-	-	-	-
M3T	0.9035	0.7521	-	-	-	-
DHO Based	-	-	-	-	-	-

Table 3: Results and Comparison of Different Models with subject-wise splitting

Model Name	Accuracy
Modified AXIAL (Ours)	0.85
Dual CNN: Multi View (Ours)	0.84
AXIAL	0.856
Majority Voting (2D)(VGG16)	0.804
Attention Transformer (TransformerConv3D)(VGG16)	0.826
AwareNet Diagnosis (AwareNet)(3D)	0.832
Attention-Guided Majority Voting (2D)(VGG16)	0.843
Majority Voting 3D (3D)(VGG16)	0.836

*: The results marked with an asterisk are considered inaccurate due to data leakage during model training and(or) evaluation. Specifically, some test samples were inadvertently exposed to the model during training, leading to inflated performance metrics. These results are included here only for comparative completeness and should not be interpreted as reliable.

** : Please interpret the Macro F1 score in these cases as the Overall F1 score, computed across all classes.

: For this model, the reported value is Matthews Correlation Coefficient (MCC) instead of Macro F1.

5 Results

Table 2 provides a detailed comparison of various deep learning models evaluated for Alzheimer’s disease classification. The results are reported in terms of training, validation,

and test accuracies, along with F1-scores for each class (AD and CN) and the overall Macro F1 score.

Our proposed **Modified AXIAL** architecture achieved the best overall performance, attaining a test accuracy of **97%** and a Macro

F1 score of **0.965**. Similarly, the **Dual CNN: Multi View (Ours)** model yielded a comparable test accuracy of **96.67%** with the highest Macro F1 score of **0.9667**, indicating its superior ability to handle multiview spatial context. Both models also maintained high F1-scores across the AD and CN classes, reflecting a balanced classification capability.

In comparison, existing models such as **Dual CNN (Single View)** and **AXIAL** achieved lower test accuracies of **95%** and **85.6%**, respectively. Notably, while the **AXIAL** model's Macro F1 score is reported as **0.712**, it actually represents the *Matthews Correlation Coefficient (MCC)* and is therefore not directly comparable with other F1-based metrics. Similarly, **ADNet3D** reports a Macro F1 score of **0.708**, though this reflects an *Overall F1 Score* rather than class-balanced F1.

Transfer learning baselines such as **ResNet**, **VGG19**, and **AlexNet** show unusually high or inconsistent performance—e.g., ResNet reports a test accuracy of **96.32%**, while VGG19 reports **93.49%**. However, these entries are marked with an asterisk (*) due to **data leakage** issues: some test samples were unintentionally present in the training set during fine-tuning, inflating the reported performance. These results are retained for completeness but are not considered reliable indicators of true generalization ability.

Other standard CNNs such as **Dual 3D CNN** and **3D CNN (Pretrained)** exhibit limited performance improvements, with test accuracies of **80.87%** and **48.45%**, respectively. Shallow CNN architectures like **AlexNet** and **LeNet** performed significantly worse on this complex classification task, with macro F1 scores of just **0.315** and **0.370**.

Transformer-based models like **ViT** also failed to generalize effectively with a validation accuracy of only **58.38%**, while advanced architectural designs such as **MBConv with Grid Attention** and **Tri-Plane Feature Fusion** provided moderate results but fell short of the proposed methods' robustness.

As mentioned above as well, after imple-

menting **Subject ID** based splitting the accuracies for AXIAL and Dual CNN were reported as 0.85 and 0.83 respectively.

In conclusion, our models, especially the Modified AXIAL and Dual CNN (Multi View), demonstrate strong discriminative ability and generalization and that ends the binary classification problem for this project.

6 Future Work

6.1 sMCI vs pMCI

MCI (Mild Cognitive Impairment) is a clinical diagnosis representing a noticeable decline in cognitive abilities. Medically, It is often considered an intermediate stge between normal aging and dementia, especially Alzheimer's Disease. It can be further classified into two subtypes:

- **sMCI (Stable Mild Cognitive Impairment):** In such cases, cognitive function remains relatively unchanged over time and there is no specific progression to dementia.
- **pMCI (Progressive Mild Cognitive Impairment):** In such cases, individuals progress to Alzheimer's disease or another form of dementia. This is characterized by increasing cognitive decline over time.

Now, that our models have achieved considerable success in differentiating between AD and CN, we moved on to possibly training the model on sMCI vs pMCI. For this we decided to employ unsupervised learning approach to analyze the MRI scans within the ADNI dataset labelled as Mild Cognitive Impairment (MCI).

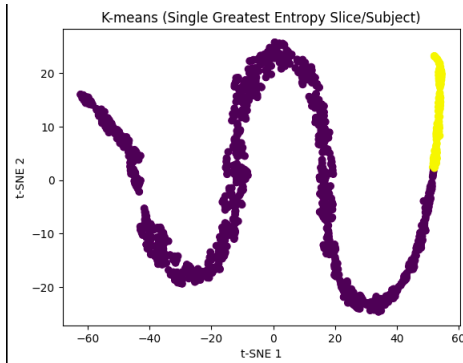
6.1.1 K-Means Clustering

The preprocessing pipeline began with metadata filtering to isolate MCI labelled scans, amounting to 1113 scans. From each MRI scan, the slice with the highest Shannon

entropy (computed from normalized intensity histogram) is selected, assuming that higher entropy slices contain more informative anatomic variability.

Each maximum entropy slice is then processed to generate a six element vector consisting of mean, standard deviation, variance, medium, minimum and maximum in terms of its pixel intensities. Now, k-means clustering is performed on the 1113 vectors with $k=2$ under the hypothesis that MCI will cluster into the abovementioned two subgroups.

K-Means Clustering achieved a **Silhouette Score of 0.759**. The silhouette score is a metric used to evaluate the quality of clusters produced and it ranges from $[-1,1]$. Hence, it is apt to conclude that the MCI scan slices are indeed converging well into two different clusters. However the issue lies in the fact that since the data is unlabelled beyond the coarse MCI category, categorizing each cluster would only be speculative right now. Dimensionality reduction was performed using t-SNE for visualisation, the result of which can be seen in the plot.



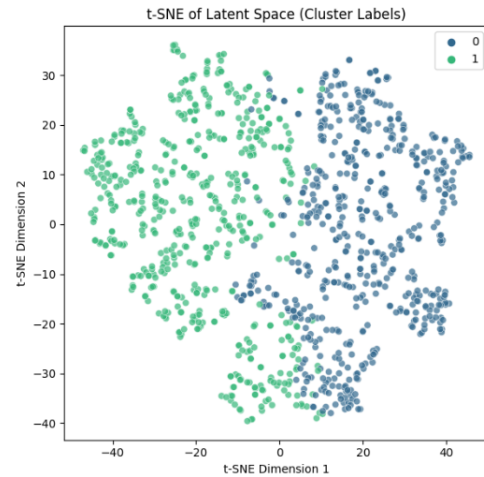
6.1.2 Autoencoder + K-Means Clustering

In the preprocessing, the central scan from each MRI scan is selected and resized into 64×64 .

The scan is passed through a symmetric autoencoder consisting of 3 convolutional layers in the encoder and 3 transpose convolutional layers in its decoder. The Autoencoder was trained for 30 epochs. The learning rate, batch

size and optimizer used for training were 0.01, 32 and Adam respectively. The final reconstruction loss was 0.039.

After the autoencoder training was complete, k-means clustering was applied on the latent space. The final internal clustering metrics were: Silhouette Score (Internal) : 0.244, Davies-Bouldin Index Index (Internal) : 1.5318.



Clearly, this section of our research is still fairly underdeveloped. We aim to devote greater attention to this and possibly combine our unsupervised clustering approaches with our trained AD vs CN model. This might be logical since the cluster with MRI scans lying closer to AD than CN would belong to pMCI and the cluster relatively closer to CN would be classified as sMCI.

6.2 Biomarkers in the ADNI Dataset

The ADNI Dataset also had biomarker data on the patients whose MRI scans our models were trained on. This data included :

- **Parental History Variables:** Parents' vital status (alive or dead), age and dementia status
- **Genetic Biomarkers:** Tr.Allele1 and Tr.Allele2 (Transformed Allele Cells), Fc.Allele1 and Fc.Allele2 (Final Call Allele Cells), Chromosome Allocation and

Raw FLuorescence intensity values from genotyping

- **Quality Scores:** GT score, Cluster Separation and GC score We want to focus on whether these biomarkers could be utilized for :
 - Train a multimodal classification model as a successor to our current AD vs CN model
 - Gain insights about how Alzheimer's Disease depends on patient biomarkers by utilising our existing models classification capabilities

6.3 Vision Transformers

Vision Transformers are computationally expensive to train. We had briefly tried implementing them but the hardware available on Kaggle's free tier isn't able to support these large models. However, we are hopeful of the results they could bring but currently do not have enough resources to implement them. Furthermore, we want to integrate cross-attention modules in our models by integrating the biomarkers mentioned above.

6.4 Interpretation of Model Prediction

Alzheimer's disease follows a predictable anatomical progression, beginning in the hippocampus. Hence, memory formation fails first while distant memories remain intact. The disease systematically spreads through the brain in five distinct stages. As the disease progresses to left hemisphere language areas, patients experience word-finding difficulties, while frontal lobe involvement impairs decision-making and planning capabilities. Visual system damage affects face and object recognition, and amygdala involvement occurs later.

Our goal is to now interpret our model predictions based on what regions of the MRI scan slices, it is focusing on majorly to assign the classification scores. The methods we aim to implement for this are :

- **GradCAM (Gradient weighed Class Activation Mapping):** It calculates gradients from the final prediction back till the target convolutional layer to create attention based heatmaps. When we run GradCAM on our model, then based on existing medical research, it should ideally highlight the hippocampus, entorhinal cortex and temporal regions in the cases of AD.
- **Saliency Map:** It measures how much each pixel affects the final classification decision. Saliency maps backpropagate till the input image itself and study how the model prediction would change if individual pixel intensities were tweaked.
- **Layer-wise Relevance Propagation:** It traces the prediction backwards through each network layer, distributing relevance scores to show contribution paths. It shows detailed attribution maps, which reveal what brain region was given importance in prediction. Instead of computing gradients, it uses a relevance conservation rule.

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